By: Wayne Yamomo

Algorithm for Cash Register:

Step 2: Print the name and developer of the program

Step 3: Print the catalogue and the prices of each item

Step 4: Input the choice of item

Step 5:

Case choice 1: print the description and price of RADIANT ROSE and total bill, and assign bill = 1250

Case choice 2: print the description and price of CELESTIAL CARNATIONS and total bill, and assign bill = 550

Case choice 3: print the description and price of SUCCINCT STARGAZER and total bill, and assign bill = 450

Case choice 4: print the description and price of ASTRAL ALSTROEMERIA and total bill, and assign bill = 350

Case choice 5: print the description and price of TRANSCEDENT TULIPS and total bill, and assign bill = 400

Case choice 6: print the description and price of DIVINE DAFFODILS and total bill, and assign bill = 650

Case choice 7: print the description and price of CHARMING CHRYSANTHEMUMS and total bill, and assign bill = 700

Case choice 8: print the description and price of ORNATE ORCHIDS and total bill, and assign bill = 1000

Case default: print wrong input and terminate the program

Step 6: Input the payment

Step 7: Print the payment received

Step 8:

While payment < bill:

Assign deficit = bill – payment

Assign bill = deficit

Input payment

Print payment received

Step 9:

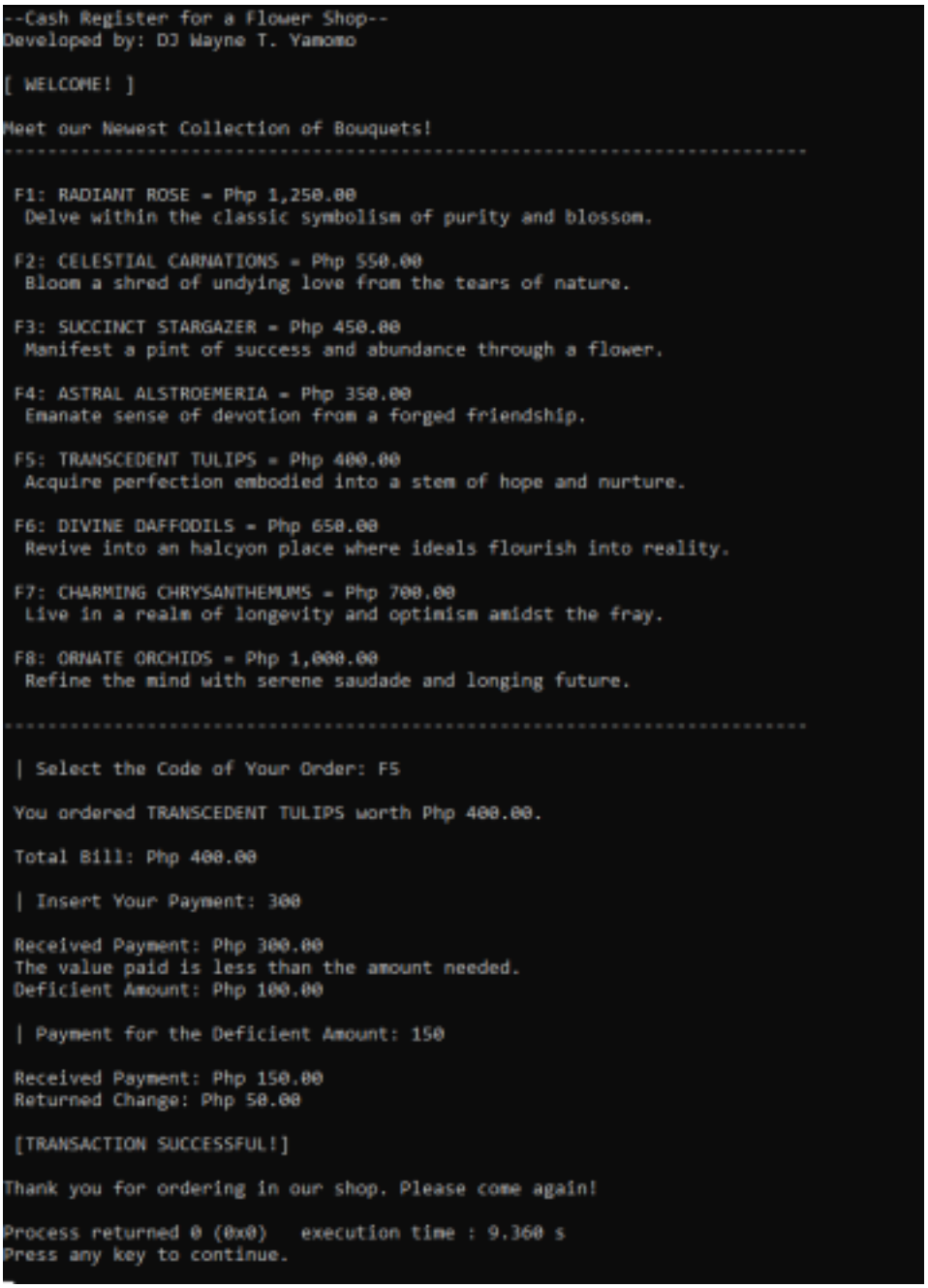
If payment > bill, then assign change = payment – bill and Print the change.

If payment == bill, then Print the "The value paid is exactly the amount needed for the bill."

Step 10: Print successful transaction of program

Step 11: Terminate the program

Output:



Algorithm for Area:

Step 1: Declare the functions

Step 2: Initialize the area variables to 0

Step 3: Print the name and developer of the program

Step 4: Call compute_areaTriangle() and assign its value to area_triangle

Initialize all variables 0

Input base and height

Calculate area_T = (base * height) / 2

Return area_T

Step 5: Print area_triangle

Step 6: Call compute_areaSquare() and assign its value to area_square

Initialize all variables 0

Input side

Calculate area_S = side * side

Return area_S

Step 7: Print area_square

Step 8: Call compute_areaRect() and assign its value to area_rect

Initialize all variables 0

Input length and width

Calculate area_R = length * width

Return area_R

Step 9: Print area_rect

Step 10: Call compute_areaCircle() and assign its value to area_circle

Initialize radius and area_C to 0

Initialize constant pi to 3.14159

Input radius

Calculate area_C = pi * (radius * radius)

Return area_C

Step 11: Print area_circle

Step 12: Terminate the program

```
--Area of Shapes Using Functions--
Developed by: DJ Wayne T. Yamomo

[ FINDING THE AREA OF A TRIANGLE ]
 | Base: 5
 | Height: 3
 The area of the given triangle is: 7.5

[ FINDING THE AREA OF A SQUARE ]
 | Side: 5
 The area of the given square is: 25

[ FINDING THE AREA OF A RECTANGLE ]
 | Length: 5
 | Width: 3
 The area of the given rectangle is: 15

[ FINDING THE AREA OF A CIRCLE ]
 | Radius: 5
 The area of the given circle is: 78.5397

Thank you for using the program!

Process returned 0 (0x0)   execution time : 9.066 s
Press any key to continue.
```